# MATH X314 – Data Visualisation

## Assessment 4B: Project Portfolio

## Documentation of Construction and Design

**Has Australia's Three-Decade safety Success Ended? The reversal story – who, where and why?**

**Submitted by**

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The document is created to record the design and construction techniques used in building the Australian road fatalities project.

## Project context and Intent

I did a lot of research before selecting the topic road safety, but this topic looks interesting and every year many people lose their lives on Australian roads. If I could find something good from visualisations and hidden patterns of road fatalities, it could help raise awareness in the Australian public.

I selected this topic because of data is coming from valid government source Australian Road Death Database (ARDD. The dataset is large and contains 58,389 rows of data from 1989 to February 2026. It consists of all important info related to road fatalities.

I started working on this because three decades of declining road fatalities suddenly started climbing up since 2020. This unusual pattern made me to work on this project. My goal was to find the reason for incremental trend since 2020 by finding answers for simple questions who, when, where and why. The focus is to explore the data and find reasons for reversal trend and reach the story to the Australian public, police, road safety officials and higher government officials.

## Changes from the proposal

My initial plan was to use every variable from the data set and create many graphic types. After continuous research, I changed my plan to 9 main graphics from 15 which made the story stronger. This change reduced redundancy while covered important variables of road fatalities with different graphics and it allowed me to reach answers for who, when, where and why. I also added the necessary interactive elements to the static plot and created the dashboard from public perspective using shiny, ggplot2 and plotly. Based on feedback from assignments, I have changed my style of story telling from formal to narrative.

## Data Selection and Preparation

The main data source is the Australian Road Death Database (ARDD). It records every fatal crash across all states in Australia. The data set is large containing 58,389 records of road fatalities from 1989 to February 2025 with important fatality information such as date, time, crash type, gender, age,

road user type and speed limits – all the necessary information needed to address the reversal rise – who, when and where. Additionally, I created necessary categorical variable like time zones using time, seasons from months to support temporal patterns. I have done data cleaning steps replacing – “Unknown”,”-9” with NA, trimming the spaces of text columns. Changed the datatype of few required columns to numeric and state population for state per capita comparison in lollipop chart came from Australian Bureau of Statistics (ABS).

## Visualisation and Design Decisions

The different chart types used in road fatalities project not just to show variability but for clean representation of each insight. The story started with line graph for 36-years trend analysis from 1989-2025 with dashed line at 2020 marking the reversal trend start point. Next, fatalities split by road users with multi-line graph to show the difference between protected and vulnerable users. To show how big is the difference used horizontal bar chart. This horizontal orientation makes road user labels readable.
A lollipop chart was used to compare state per capita rates – it makes thin bars and highlight the outliers and adds diversity. Heat map is used to observe the temporal patterns of time, day and seasons – darker tile colours represent risky. Stacked bar chart for crash types to show the difference used colour hue for single and multi-crash types.

Finally, story ends  with box plot to show the distributions of road fatalities by demographics.To show difference between categories in bar graph used, different colours assigned to each category.
Overall followed simple techniques to avoid complexity in visualisations.

## Narrative and Storytelling design

The story is divided into 3 sections; each section adds value to the previous sections. First section focuses on the national trend, next on when and where fatalities are higher and the third on who are affecting more and why.

Each section is designed a way to provide solution for reversal trend by answering – who, where and when. I changed my narrative style from formal (“Road Fatalities trend in Australia”) to narrative completely (“Has three-decade success of road safety ended in Australia”). Additionally, I changed the titles and subtitles from formal insights to engaging questions to create eagerness in audience and answering them through subtitles.

## Technical implementation

The project is completely built in R language, using R-Studio. I used a.qmd file for writing code and report, used core packages in building project – read excel  for importing data, tidy verse for data cleaning, crosstalk and plotly for adding interactive elements in the graphs, ggplot2 for creating plots, shiny dashboard for dashboard layout, scales for number formatting, shiny for dashboard creation.
I faced many challenges while doing the project such as name conflicts, chunk break downs, and syntax errors and  adjusting the text.

## Reflection on Project Development

I learnt many projects while doing the project. At first, I thought creating visualisations and comparing the variables is the main thing over time I understood that random comparisons and plotting graphics doesn't bring much value.
My focus is completely shifted towards reversal trend after observing reversal trend in this project. Then I tried to find from different perspectives then I got to know vulnerable users are the main for this reversal trend.

Adding interactive additionally added value to my project as I got opportunity to compare more combinations and got to know the solution finally for -who, when, where and why.